LINUX WEB SERVICES PRACTICAL ASSIGNMENT

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Task Name: Apache, PHP, FTP, Node.js and Directory Access Control

Platform: RHEL 9 / Rocky Linux / AlmaLinux commands. Use sudo where required. Replace SERVER_IP and FTPUSER with your actual values.

Safety note: Keep this lab on a trusted network. Basic FTP is unencrypted; use SFTP for real administration. Do not expose lab services to the public internet.

Task 1 — Install Apache and PHP; create index.php

Question: Install Apache and PHP and display “Welcome to My Foodie Blog” in a browser.

```
sudo dnf install -y httpd php
sudo systemctl enable --now httpd
sudo systemctl status httpd

sudo tee /var/www/html/index.php > /dev/null <<'EOF'
<?php
echo "Welcome to My Foodie Blog";
?>
EOF

sudo firewall-cmd --permanent --add-service=http
sudo firewall-cmd --reload
sudo apachectl configtest
```

Test: Open http://127.0.0.1/ on the server, or http://SERVER_IP/ from another computer. Expected page text: **Welcome to My Foodie Blog**. If PHP source is shown instead of the message, fix PHP integration before continuing.

Task 2 — Configure a custom 404 error page

Question: Show the message “Oops! This page is as empty as your cart.” when a page is missing.

```
sudo tee /var/www/html/404.html > /dev/null <<'EOF'
<!doctype html>
<html><head><title>Page not found</title></head>
<body>
<h1>Oops! This page is as empty as your cart.</h1>
<p>We could not find the page you requested.</p>
<a href="/">Return to the Foodie Blog</a>
</body></html>
EOF

sudo tee /etc/httpd/conf.d/custom-errors.conf > /dev/null <<'EOF'
ErrorDocument 404 /404.html
EOF

sudo apachectl configtest
sudo systemctl reload httpd
```

Test: Visit http://127.0.0.1/this-page-does-not-exist. Expected: the custom cart message. Apache configtest should return **Syntax OK**.

Task 3 — Install FTP server and upload a sample image

Question: Set up vsftpd and upload an image into the website directory.

```
sudo dnf install -y vsftpd
sudo cp -a /etc/vsftpd/vsftpd.conf /etc/vsftpd/vsftpd.conf.bak
sudo vi /etc/vsftpd/vsftpd.conf
```

Ensure these settings are present once and not contradicted by duplicate active lines:

```
anonymous_enable=NO
local_enable=YES
write_enable=YES
local_umask=022
```

```
sudo systemctl enable --now vsftpd
sudo systemctl restart vsftpd
sudo systemctl status vsftpd
```

In FileZilla or another FTP client, connect to the server IP on port 21 using a lab Linux account. Upload a sample image (for example, food.jpg) into that user's home directory. Then copy it to the web root:

```
sudo install -d -m 755 /var/www/html/images
sudo install -m 644 /home/FTPUSER/food.jpg /var/www/html/images/food.jpg
sudo firewall-cmd --permanent --add-service=ftp
```

```
sudo firewall-cmd --reload
```

Replace FTPUSER and food.jpg with the real username and filename. Remote FTP passive mode may require a configured passive port range. Test http://127.0.0.1/images/food.jpg. Do not enable anonymous uploads.

Task 4 — Run a Node.js playlist JSON API on port 3000

Question: Run a separate Node.js service that returns a mock “Now Playing” playlist.

```
sudo dnf install -y nodejs
command -v node
sudo mkdir -p /opt/now-playing
sudo tee /opt/now-playing/server.js > /dev/null <<'EOF'
const http = require('http');
const playlist = {
  service: 'Now Playing',
  playlist: [
    { title: 'Morning Coffee', artist: 'The Sample Band' },
    { title: 'City Lights', artist: 'Demo Artist' },
    { title: 'Weekend Drive', artist: 'The Testers' }
  ]
};
http.createServer((req, res) => {
  if (req.url === '/api/playlist') {
    res.writeHead(200, {'Content-Type': 'application/json; charset=utf-8'});
    res.end(JSON.stringify(playlist, null, 2));
  } else {
    res.writeHead(404, {'Content-Type': 'application/json'});
    res.end(JSON.stringify({error: 'Not found'}));
  }
}).listen(3000, '0.0.0.0', () => console.log('Playlist API on port 3000'));
EOF
```

Create a systemd service so it runs in the background. If command -v node reports a path other than /usr/bin/node, use that path in ExecStart.

```
sudo tee /etc/systemd/system/now-playing.service > /dev/null <<'EOF'
[Unit]
Description=Demo Now Playing JSON API
After=network.target
[Service]
Type=simple
WorkingDirectory=/opt/now-playing
ExecStart=/usr/bin/node /opt/now-playing/server.js
Restart=on-failure
User=nobody
Group=nobody
[Install]
WantedBy=multi-user.target
EOF
sudo systemctl daemon-reload
sudo systemctl enable --now now-playing
sudo systemctl status now-playing
```

For remote access only, open port 3000 on the lab firewall:

```
sudo firewall-cmd --permanent --add-port=3000/tcp
sudo firewall-cmd --reload
```

Test: Visit http://127.0.0.1:3000/api/playlist. Expected: JSON with the service name and three sample tracks. Also test the PHP page at http://SERVER_IP/. Do not open port 3000 if only local testing is needed.

Task 5 — Restrict /secret to localhost (127.0.0.1)

Question: Allow only requests from localhost to access /var/www/html/secret.

```
sudo mkdir -p /var/www/html/secret
echo 'Localhost-only secret area' | sudo tee /var/www/html/secret/index.html
sudo chmod 755 /var/www/html/secret
sudo chmod 644 /var/www/html/secret/index.html

sudo tee /etc/httpd/conf.d/secret-localhost.conf > /dev/null <<'EOF'
<Directory "/var/www/html/secret">
    Require ip 127.0.0.1
</Directory>
EOF

sudo apachectl configtest
sudo systemctl reload httpd
```

Test on the server:

```
curl -i http://127.0.0.1/secret/
```

Expected locally: HTTP 200 and “Localhost-only secret area”. From another computer, visit http://SERVER_IP/secret/; expected HTTP 403 Forbidden. This rule allows IPv4 localhost 127.0.0.1, not IPv6 ::1.

Troubleshooting

• Apache service or config error: sudo apachectl configtest and sudo journalctl -u httpd -xe

• Check server IP: ip addr; check firewall: sudo firewall-cmd --list-all

• Node.js service: sudo systemctl status now-playing; sudo journalctl -u now-playing -e; ss -lntp | grep 3000

• FTP upload failure: verify vsftpd is running, local-user settings, credentials, and write permissions.

• Custom 404 not shown: verify /var/www/html/404.html and /etc/httpd/conf.d/custom-errors.conf, then reload Apache.

Verification checklist

Task	Test	Expected result
PHP homepage	http://127.0.0.1/	Welcome to My Foodie Blog
404 page	/this-page-does-not-exist	Custom cart message
Uploaded image	/images/food.jpg	Image displays
Node.js API	http://127.0.0.1:3000/api/playlist	JSON playlist
Secret folder	curl -i http://127.0.0.1/secret/	Local 200; remote 403

Short viva questions and answers

1. What is Apache? A web server that receives HTTP requests and serves pages.

2. What does PHP do? It executes server-side code and sends the result to the browser.

3. What does ErrorDocument 404 do? It specifies the page shown when a requested URL is not found.

4. What is vsftpd? A Linux FTP server; standard FTP is unencrypted, so SFTP is safer for real use.

5. Why use port 3000? It allows the Node.js service to run separately from Apache on port 80.

6. What does Require ip 127.0.0.1 mean? Only requests from IPv4 loopback are permitted for that directory.

7. Why run apachectl configtest? It checks Apache configuration syntax before reloading.

Submission note: Run the commands on your own Linux system and add actual screenshots or outputs. Do not mark a task successful until it has been tested.